Natural Language Processing

https://web.stanford.edu/~jurafsky/slp3/

Viviana Mascardi

Following the well known textbook by Dan Jurafsky and James H. Martin on Speech and Language Processing, https://web.stanford.edu/~jurafsky/slp3/, the ***Natural Language Processing*** *course devotes a large share of lessons to "Annotating Linguistic Structure" (Volume III), extended with lessons on the birth of human language, its semantics, and its pragmatics; the remaining classes are devoted to "Large Language Models" (Volume I). Contents of Volume I that have already been dealt with in other Machine Learning courses (Logistic Regression, Neural Networks, RNNs, etc...) are not covered.*

The exam consists of a written part and an individual project, that can be proposed by the students themselves, to leave them free to experiment in a practical way those topics that they liked more.

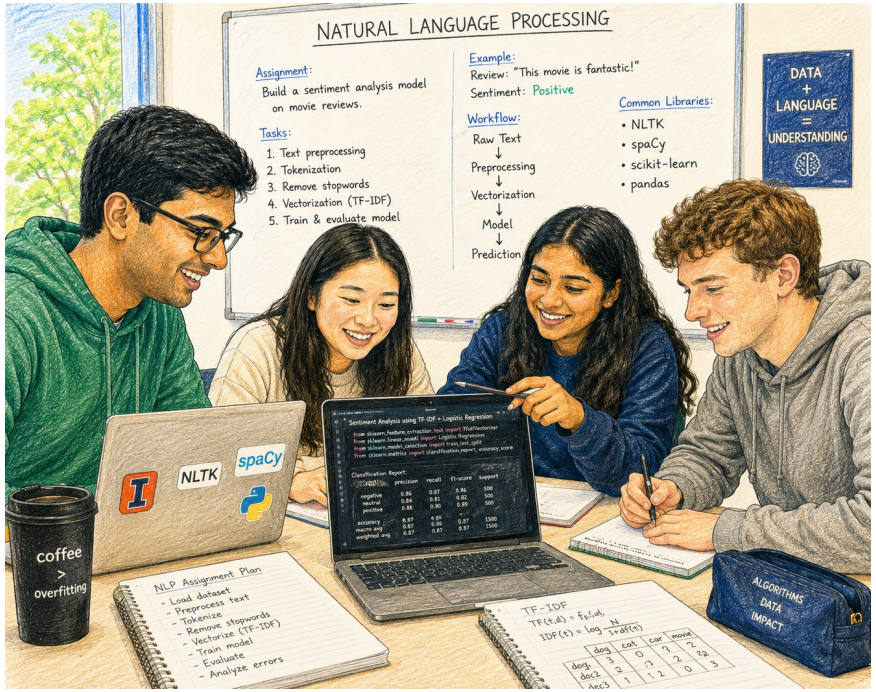


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Natural Language Processing

Program

NLP 2026-2027: this is just a preliminary draft, not the final schedule

Wednesday 11-13 (we start at 11.15 sharp)
Thursday 14-16 (we start at 14.00 sharp)
always in room 218

1. 23/09 VM: Introduction to the NLP course. Some thoughts on the birth of language. Terminology; Syntax: RegEx (with LAB)
2. 24/09 VM: Syntax: Word Level: segmentation, stop words, TF-IDF, Stemming, Lemmatization (with LAB)
Please note that there will be no lesson on September 30 for teachers' commitments that cannot be moved
3. 01/10 VM: Syntax: Minimum Edit Distance, Sentence Level, POS Tagging with deductive (rule-based) approaches (with LAB)
4. 07/10 VM: Syntax: Sentence Level, More on Python libraries for managing grammars and POS tagging; POS Tagging with inductive (Hidden Markov Models) approaches (with LAB)
5. 08/10 VM: Semantics: Distributional semantics, Frame Semantics, Model-theoretic semantics, Lexical Semantics

Natural Language Processing

6. 21/10 VM: Semantics: A reasoned comparison among different types of semantics
7. 22/10 VM Semantics: Ontologies and the Semantic Web, Applications
8. 28/10 Protégé, BabelNet, libraries for playing with NLP, text summarization (LAB by Andrea Gatti)
9. 29/10 Chatbots and Agentic AI (LAB by Andrea Gatti)
10. 04/11 VM: Ontologies wrap up, Syntax + Semantics = Text features
11. 05/11 VM: Pragmatics
12. 11/11 VM: Wrap up on the classical NLP approaches
13. 18/11 GM: Bert, LLMs, GPT
14. 19/11 GM: Bert, LLMs, GPT
15. 25/11 GM: Bert, LLMs, GPT
16. 26/11 GM: Bert, LLMs, GPT

Optional lessons: in some slot in early december we will fix one day for exercices in preparation of the exam, and one day for informal feedback on project proposals.

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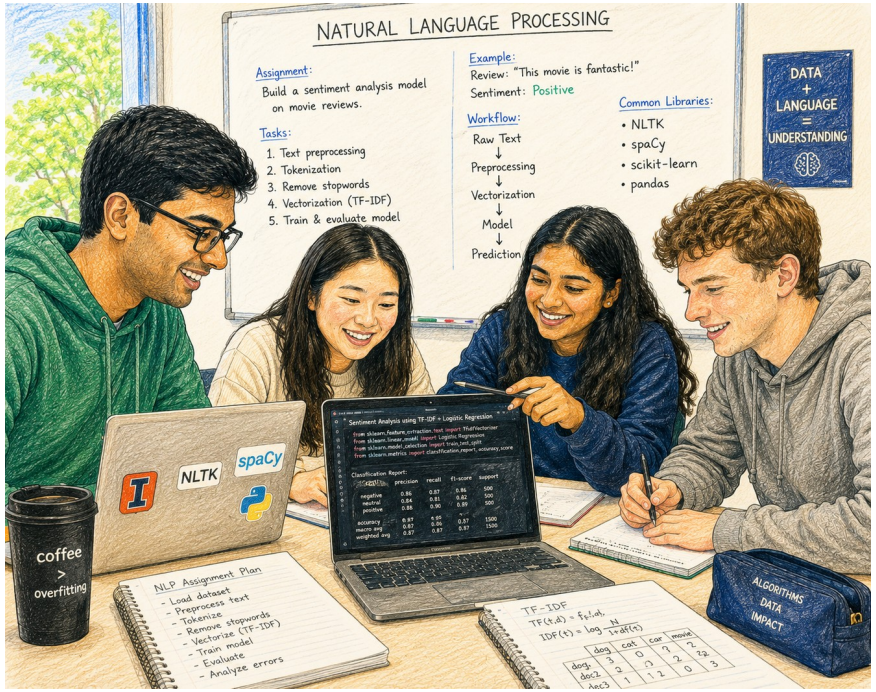


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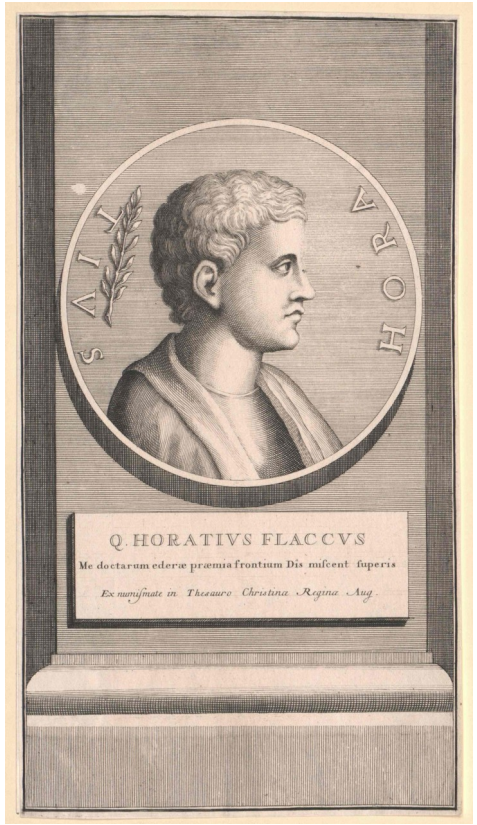
What does characterize human language w.r.t. communication among animals?

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Suggestion 1 (this is really difficult to catch!!!)





What does characterize human language w.r.t. communication among animals?

Suggestion 2: bees' language represents an exception to the most evident difference between human language and animal's communication, with bees being able to convey a specific piece of information that no other animals convey in the wild...

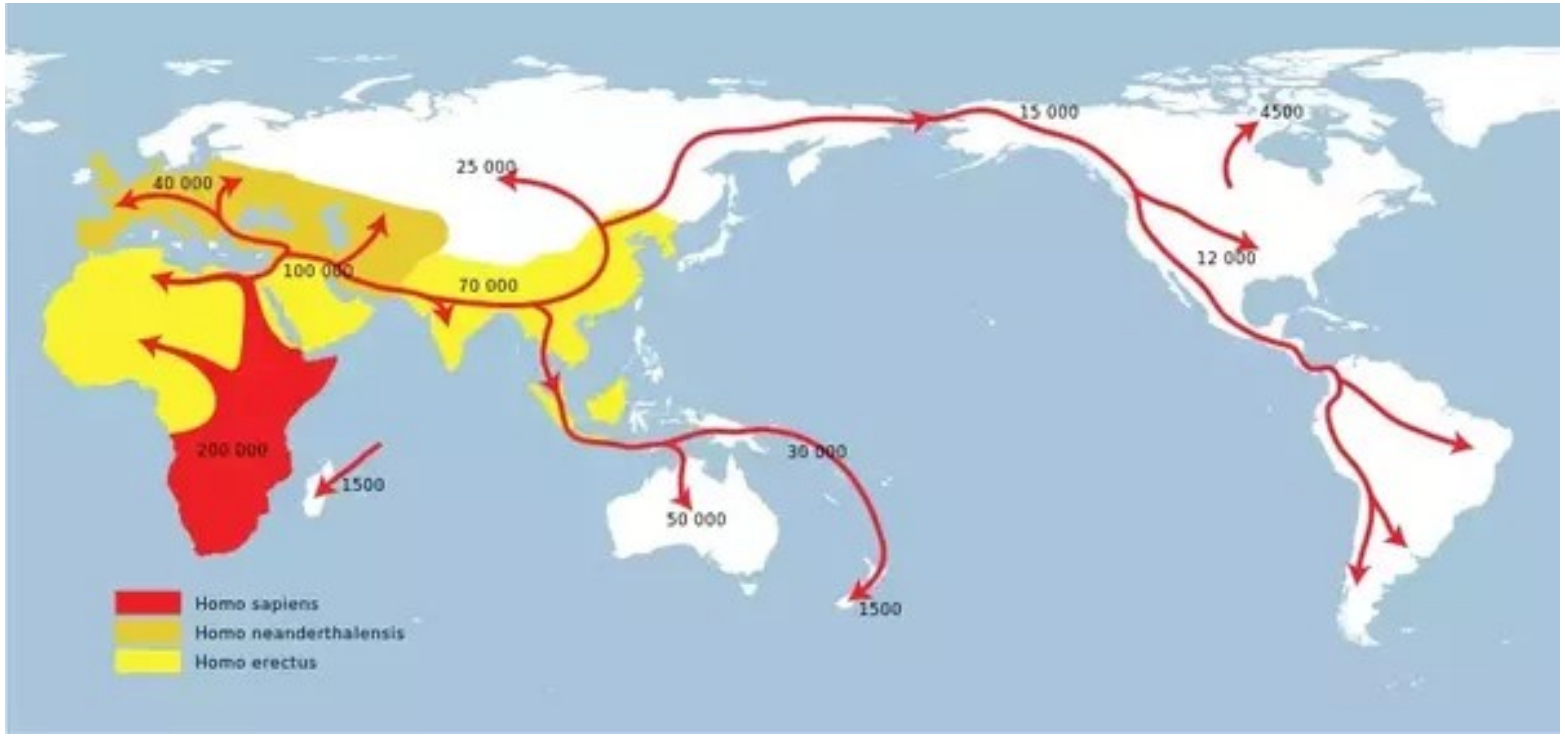


On the birth of human language

Years ago	Timeline taken from *Sapiens A Brief History of Humankind by* Yuval Noah Harari
6 MYA	Last common grandmother of humans and chimpanzees
2.5 MYA	Evolution of the genus Homo in Africa [maybe even earlier: LD 350-1] First stone tools
2 MYA	**Humans spread from Africa to Eurasia**. Evolution of different human species. Homo Rudolfensis and Homo Ergaster evolve in Africa Homo Erectus evolves in East Asia
500,000 YA	Neanderthals evolve in Europe and the Middle East
200,000 YA	Homo sapiens evolves in East Africa
70,000 YA	**Sapiens spreads out of Africa**
70,000-30,000 YA	Cognitive revolution: invention of boats, arrows, needles. Beginnings of art, religion, commerce, social stratification, and language.
45,000 YA	Sapiens settle Australia
30,000 YA	Extinction of Neanderthals
16,000 YA	Sapiens settle America
13,000 YA	Extinction of Homo Floresiensis **Homo Sapiens only surviving human species**
12,000 YA	Agricultural revolution Permanent Settlements

Adapted from Harari's time line of the evolution of humankind in his 2014 book Sapiens A Brief History of Humankind

On the birth of human language



Movement of Homo Sapiens through the world. Image courtesy Wikimedia Commons
Downloaded from https://www.lizatlancaster.co.za/blog/human-ancestors-part-ii/

On the birth of human language

Timeline and Archaeological Milestones

135,000 Years Ago: Genomic and geographic data suggest the biological and cognitive capacity for language was already fundamentally present.

100,000 Years Ago: Archaeological records show an explosion of symbolic behaviors like art and complex tools, indicating active social communication.

Gradual Origins: Scientists widely agree language developed from a complex mix of hand gestures, facial expressions, and vocalizations over time.

https://news.mit.edu/2025/when-did-human-language-emerge-0314

In your opinion, did the Neanderthals speak?

On the birth of human language

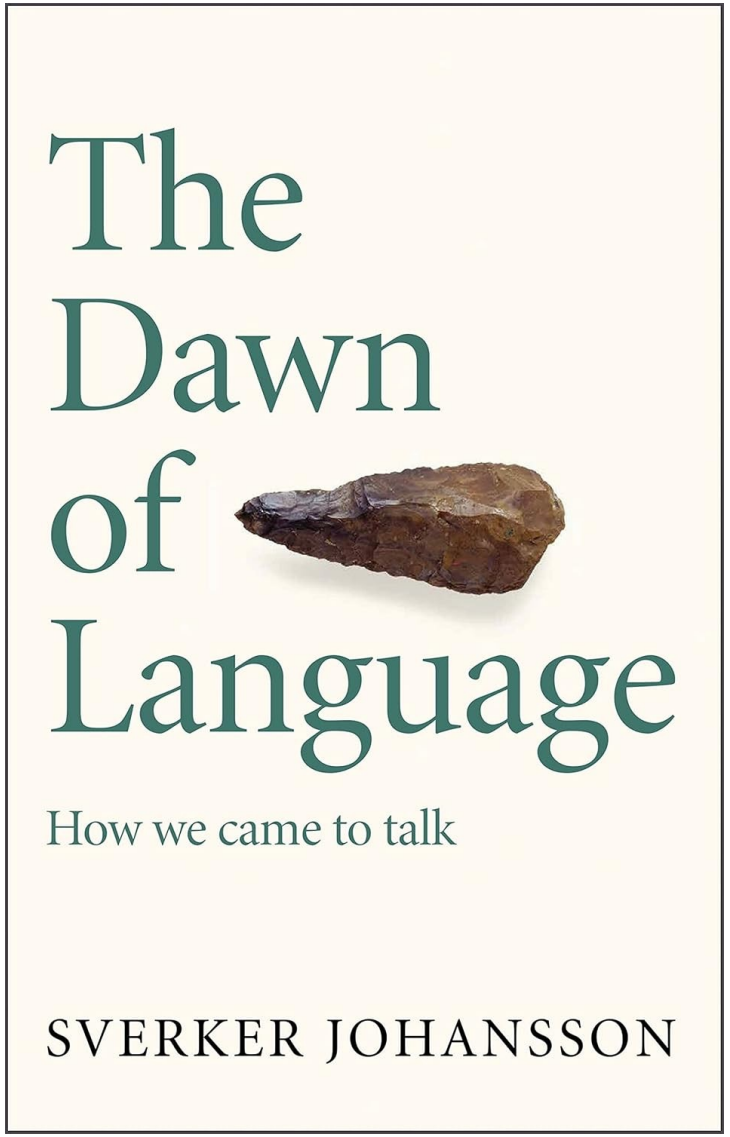
Key Drivers of Language Evolution

...according to many scientists including Johansson, one major key driver was “the obstetrical dilemma”.

Upright Walking (Bipedalism): Restructured the human pelvis into a basket-like shape, narrowing the birth canal to support internal organs and efficient locomotion.

Large Brains: Increased hominid brain size dramatically over millions of years, creating a tight physical fit between the baby and the mother's pelvis.

Consequence: labor and delivery is difficult in the human species (not only, but especially)



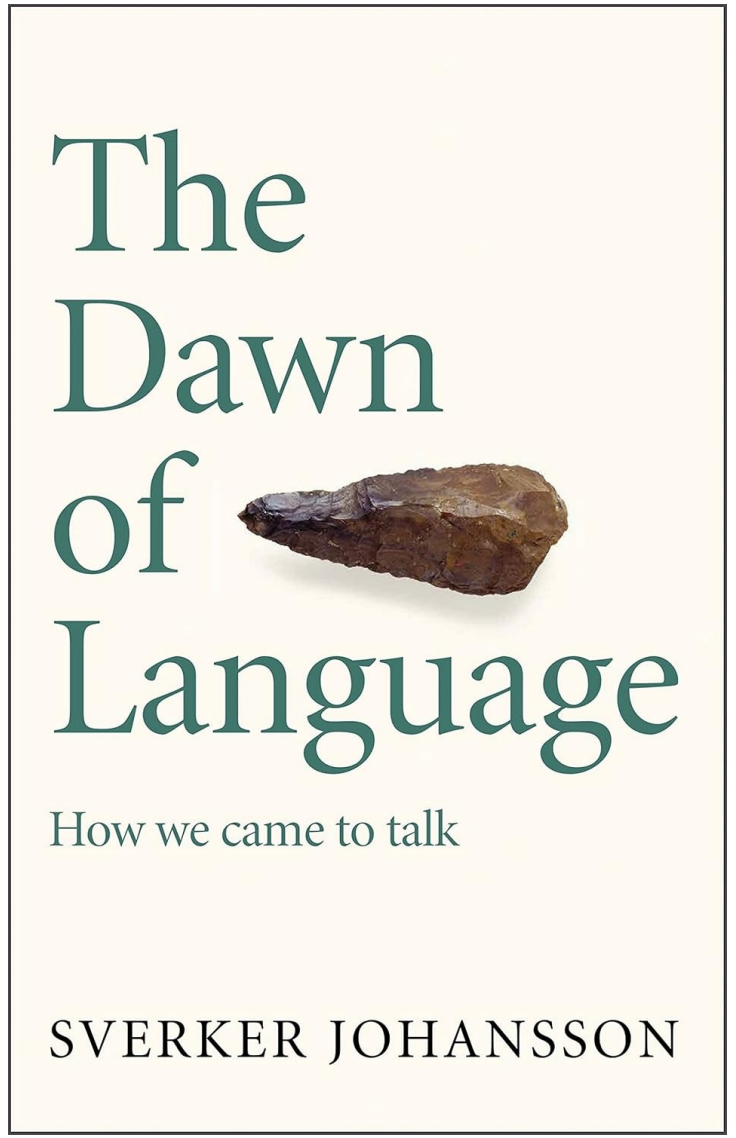
On the birth of human language

Key Drivers of Language Evolution

Also, babies are born immature: they cannot cling as all higher non human primates do...

And they are in need for protection of their family or social group for a very long time





In your opinion, how may these facts relate with the birth of language?









On the birth of human language

Evolutionary Pressure	Physical Consequence	Behavioral & Language Outcome
Bipedalism & Big Brains	Narrow birth canal; difficult, dangerous labor.	Need for **midwives and social support**, forcing cooperative social structures.
Neurological Immaturity	Babies are born completely helpless and cannot cling.	Mothers must **"put the baby down"**, using vocalizations to soothe and monitor them from afar.
Extended Childhood	Years of dependency requiring group protection.	Long childhoods provide a **massive window for learning** complex language from the community.

On the birth of human language

Key Drivers of Language Evolution

Survival and Cooperation: Early humans needed to coordinate group hunts, plan defenses against predators, and share crucial resources.

Social Bonding: Communication grew from basic physical grooming into storytelling around fires, building community trust and deep empathy.

Abstract Thought: Language enabled humans to reference things not immediately present, like the past, future, or imaginary cultural concepts.

On the birth of human language

Feature	🧬 Continuity Theory	⚡ Discontinuity Theory	🖐️ Gesture-First Theory
Core Idea	Language evolved **gradually** over millions of years out of animal communication systems.	Language appeared **suddenly** as a unique, exclusively human genetic mutation.	Language originated from **physical gestures** and manual signs before shifting to speech.
Key Mechanism	Natural selection slowly shaped the vocal tract and brain tissue to handle complex communication.	A sudden cognitive "leap" or single genetic mutation re-wired the brain for universal grammar.	Bipedalism freed the hands, allowing early humans to use detailed hand and facial signs to communicate.
Primary Driver	The gradual need for social bonding and survival coordination in expanding groups.	An internal computational upgrade in the brain, prioritizing thought organization over communication.	The mechanical advantage of visual "miming" before vocalization evolved to take over.
Famous Proponents	Charles Darwin	Noam Chomsky	Michael Corballis

The role of gossip!!!



The role of gossip!!!

Gossip acted as a vocal substitute for physical grooming in primates, allowing early humans to bond and manage social relationships in groups too large for direct touch alone.

Primate roots: Primates maintain social bonds and group cohesion through physical grooming (picking ticks and fleas).

The group size limit: As early human groups grew larger physical grooming became too time-consuming to bond every member.

Vocal grooming: Language evolved as a form of "vocal grooming" that allowed humans to bond with multiple individuals simultaneously instead of one-on-one.

The role of gossip!!!

Functions of Early Gossip

Information sharing: Transmitted crucial reputational data about who could be trusted and who was a free rider.

Social cohesion: Strengthened group identity, enforced social norms, and managed complex social networks without written records.

Conflict reduction: Alleviated internal group friction by aligning moral judgments and expectations.